Assignment 1: Panoramic Image Stitching Pipeline

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Abstract

This report explains a simple panoramic image stitching system that combines three overlapping images into one wide-view image. The pipeline uses feature detection, feature matching, homography estimation, and image warping. A blending technique is applied to remove visible seams and brightness differences. The final result is compared with a naive stitch to show the improvement.

1 Introduction

Image stitching is a computer vision technique used to combine multiple overlapping images into a single panoramic image. It is commonly used in photography, mapping, and robotics. This project implements an image stitching pipeline without using any automatic panorama tools, focusing on understanding each processing step.

2 Objective

The main goals of this assignment are:

- Capture three overlapping images
- Detect feature points using SIFT
- Match features between images
- Compute homography using RANSAC
- Warp images into a common reference frame
- Blend images to remove seams

3 Data Collection

Three images (left, center, and right) were captured using a smartphone. The camera was rotated about the same point to avoid large perspective changes. Each image overlaps with the next by about 30–40%.



Figure 1: Input images: Left, Center, and Right

4 Image Stitching Pipeline

The image stitching process follows these steps:

1. Convert images to grayscale
2. Detect features using SIFT
3. Match features between images
4. Estimate homography using RANSAC
5. Warp images to a common frame
6. Perform blending
7. Crop black borders

5 Feature Detection using SIFT

SIFT (Scale Invariant Feature Transform) detects special points in images such as corners and edges. These points remain stable even when the image is rotated, resized, or slightly illuminated differently.



Figure 2: Detected SIFT keypoints

6 Feature Matching

The detected SIFT features from different images are matched using a Brute Force matcher. Lowe's ratio test is applied to keep only strong and correct matches.

7 Homography Estimation

Homography is a matrix that maps points from one image to another. It is written as:

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = H \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

RANSAC is used to remove incorrect matches and compute a reliable homography.

8 Image Warping

The left and right images are warped into the coordinate system of the center image using the computed homography. This aligns all images into one common view.

9 Naive Stitching

In naive stitching, the warped images are directly placed on top of each other. This creates visible lines, brightness differences, and misaligned edges.



Figure 3: Naive stitched panorama

10 Blending and Seam Removal

To remove visible seams, weighted blending is applied. Pixel values are smoothly mixed in overlapping regions so that the transition between images looks natural.



Figure 4: Seamless Blended panorama

11 Final Panorama

After blending, black borders are removed by cropping. This produces a clean rectangular panorama.



Figure 5: Final stitched panorama

12 Comparison



(a) Naive Stitch

(b) Final Stitch

Figure 6: Naive vs Final panorama

13 Conclusion

The panoramic image stitching pipeline was successfully implemented using SIFT, homography, RANSAC, and blending. The final result shows much better alignment and visual quality compared to naive stitching.

14 References

- OpenCV Documentation: `https://docs.opencv.org`
- Lecture Notes
- OpenCV Python SIFT Feature Detection (SIFT Algorithm Explained + Code) `https://www.youtube.com/watch?v=flFbNka62v8`